

Why Rabbit Prothrombin Is Not a Direct Stand-In for Human Prothrombin

A magazine-style explainer on why the rabbit protein is not a direct stand-in for the human one

Main idea: rabbit prothrombin has a deletion near the first factor Xa cleavage region, and that break lands right where the human protein needs to be cut. That makes rabbit a poor stand-in for human prothrombin when the assay depends on F1.2 generation.

Built from the files in this project. Human sequence: **P00734**. Rabbit sequence used: **XP_002709128.3**. Rabbit model type: **alphafold_website_prediction**.

Report generated on June 04, 2026. The images in this PDF are listed below and explained in the sections that follow.

How to use this PDF: start with the question, then read the pictures in order. The mechanism graphic explains the chemistry, and the 3D renderings show exactly where the rabbit sequence changes the shape of the cut site.

What this study asks

This feature turns a protein sequence problem into a visual story about shape, cutting, and species differences.

It includes the main renderings created by the project and explains what each one means for prothrombin research.

What we compared

Human prothrombin and rabbit prothrombin sequences

Human AlphaFold structure and rabbit threaded visualization PDB

New cleavage mechanism illustration

Renderings, alignment, and metrics

Why it matters

To see whether they match around the first cleavage site

To show where the sequence difference sits in 3D space

To show how the cut leads to F1.2 and why the rabbit deletion matters

To make the result easy to inspect and reproduce

1. Why this study matters

Scientists often use animal proteins to study human biology. That only works when the animal protein behaves like the human one in the place that matters. Here, the important place is the first factor Xa cleavage region of prothrombin, because that is where the F1.2 fragment comes from, like a zipper opening at one exact tooth.

The original rabbit study asked whether rabbit F1.2 could be used as a sign that coagulation had been activated. The answer was basically no, because rabbit plasma did not show the same F1.2 response as human plasma. Our project checks the sequence and 3D context behind that difference, like comparing a bridge with one missing plank right where the first person would step.

2. How we did the study

We built the analysis step by step, and every file is saved in the project so another person can repeat it.

- We fetched the human prothrombin sequence from UniProt.
- We searched for rabbit prothrombin first in UniProt and then in NCBI Protein.
- We aligned the two sequences with Biopython to find the exact location of gaps and matches.
- We downloaded the human AlphaFold structure and used the rabbit model to compare the same protein in another species.
- We superimposed the rabbit structure onto the human structure and recorded metrics for the whole protein and the cleavage-site windows, so the difference could be seen as a real 3D shape change instead of just a line of letters.
- We made figures that show the domain map, the alignment, and several 3D-style renderings.

The rabbit model came from the AlphaFold website, so we could compare a real predicted rabbit structure with the human one.

Rabbit model strategy: alphafold_website_prediction. Structure metric status: complete.

Alignment snapshot

Human UniProt P00734 precursor sequence is the numbering source of truth.
Human precursor 314/315 is labeled as the first Xa site, equivalent to the mature R271 region.
Human precursor 363/364 is labeled as the second Xa site, equivalent to the mature R320 region.

human	1	MAHVRGLQLPGCLALAAALCSLVHSQHVLAPQQA	RSLLQ	RVRRANT-FLEEVRKGNLERECVEETCSYEEAFE	LESSTA	79																		
rabbit	1	MEHVPGLRLAGCLAAAAVSLVCSQQVFLAPQ	RAMSL	LQRARRANTGFLEEVRKGNLERECYEEVCSYEEAFE	LESSSA	80																		
human	80	TDVFWAKYTACETARTPRDKLAACLEGNCAEGL	TNYR	GHVNITRSGIECQLWRSRYPHKPEINSTTHPGADL	QENFCRN	159																		
rabbit	81	TEAFWAKYTACEPVRTPRENFTNCMEGHCAQGL	TNYR	QVSVTQSGIECQLWRSRYPHNPEINFTHPEADL	KENFCRN	160																		
human	160	PDSSTTGWPWCYTDDTPVRRQEC	SIPVC	GQDQVTVAMTPRSE	GSSVNLSPPLEQ	VCVPDRGQYQYGR	LAVTT	HGLPCL	AWSS	239														
rabbit	161	PDGSTTGWPWCYTRDPSV	REEC	SVPD	CSQGQVTL	ELRPRTGHSGVT	QPPPLE	PCVSR	GRHYQGR	LAVTKQ	GT	PC	LAWSS	240										
human	240	AQAKALSKHQDFNSAVQL	VENFC	RNP	DGDEEG	VWCYVAGKPGD	FGYCD	LN	YCEE	AVEEET	GDGL	DEDS	DR	AIEGR	TATSE	319								
rabbit	241	PEAQLSRGKSFSPQV	LVENF	CRNP	DGDEEG	AWCYVAGRP	SSFEY	CGLN	YCEE	IIIEE	----	VLGLEE	AIEGR	TTEQE	315									
human	320	YQTFNPNRTFGSGEAD	CLRPL	FEKKS	LEDK	TERELLES	YIDGR	IVEG	SDAE	IGMSPW	QV	MLFR	KSPQ	ELL	CGAS	LISDR	399							
rabbit	316	FQTFNQNFTFGTGEAD	CLRPL	FEKKS	LKDER	EEELLES	YIHGR	IVGG	RDAIG	SAPWQ	VM	FRKSP	QELL	CGAS	LISDR	395								
human	400	WVLTAAHCLLYPPWD	KNF	TEND	LLVR	IGKHS	RTRY	ERNIE	KISM	LEKI	YIH	PRYN	WREN	LD	RDI	AL	MK	LK	PP	VA	FS	DY	IH	479

```

      |||
rabbit 396 WVLTAACHCLLYPPWDKNFTVNDILVRIGKYARSRYERNMEKISTLEKIIHPGYNWRENLDRIALMKLKKPVAFSDYIH 475
      |||
human 480 PVCLPDRETAASLLQAGYKGRVTGWGNLKETWTANVGKGQPSVLQVVNLPIVERPVCKDSTRIRITDNMFCAQYKPDEGK 559
      |||
rabbit 476 PVCLPDQKQIVTSLQAGHKGRVTGWGNLKEMWTVNMNEVQPSVLQVMVNLPLVERPICKASTRIRVTDNMFCAGYKPEEGK 555
      |||
human 560 RGDACEGDSGGPFVMKSPFNRRWYQMGIVSWGEGCDRDGKYGFYTHVFRLLKKWIKVIDQFGE 622
      |||
rabbit 556 RGDACEGDSGGPFVMKNPYNNRWYQMGIVSWGEGCDRDGKYGFYTHVFRLLKKWIRKVMVDRFG- 617

```

3. What the pictures show

The first figure is a domain map of human prothrombin. It shows where the important parts are, including the two factor Xa cleavage sites. The second figure is the sequence alignment, where the rabbit gap near the first site is visible. The later 3D renderings zoom into that same region, where the protein shape looks like a bent metal hinge with one missing screw, so the joint no longer closes the same way.

The most important result is simple: there is a rabbit gap near the first Xa-sensitive region. That means rabbit prothrombin is not an exact copy of human prothrombin in the part that matters for this assay. In 3D, the rabbit version looks like a rope with one short section snipped out, so the nearby folds sit differently and the cutting site is not framed the same way.

Key result: the rabbit deletion sits near the human first Xa cleavage site, the same region needed to generate F1.2 in human prothrombin activation. That makes the rabbit protein a weaker match right at the spot where the action begins, and that is exactly why the original hypothesis holds up.

Structure metrics

The analysis recorded this status: complete. The rabbit model source type is `alphafold_website_prediction`. Because the rabbit model is an AlphaFold prediction, the RMSD numbers are only a shape check, not proof of real reaction speed.

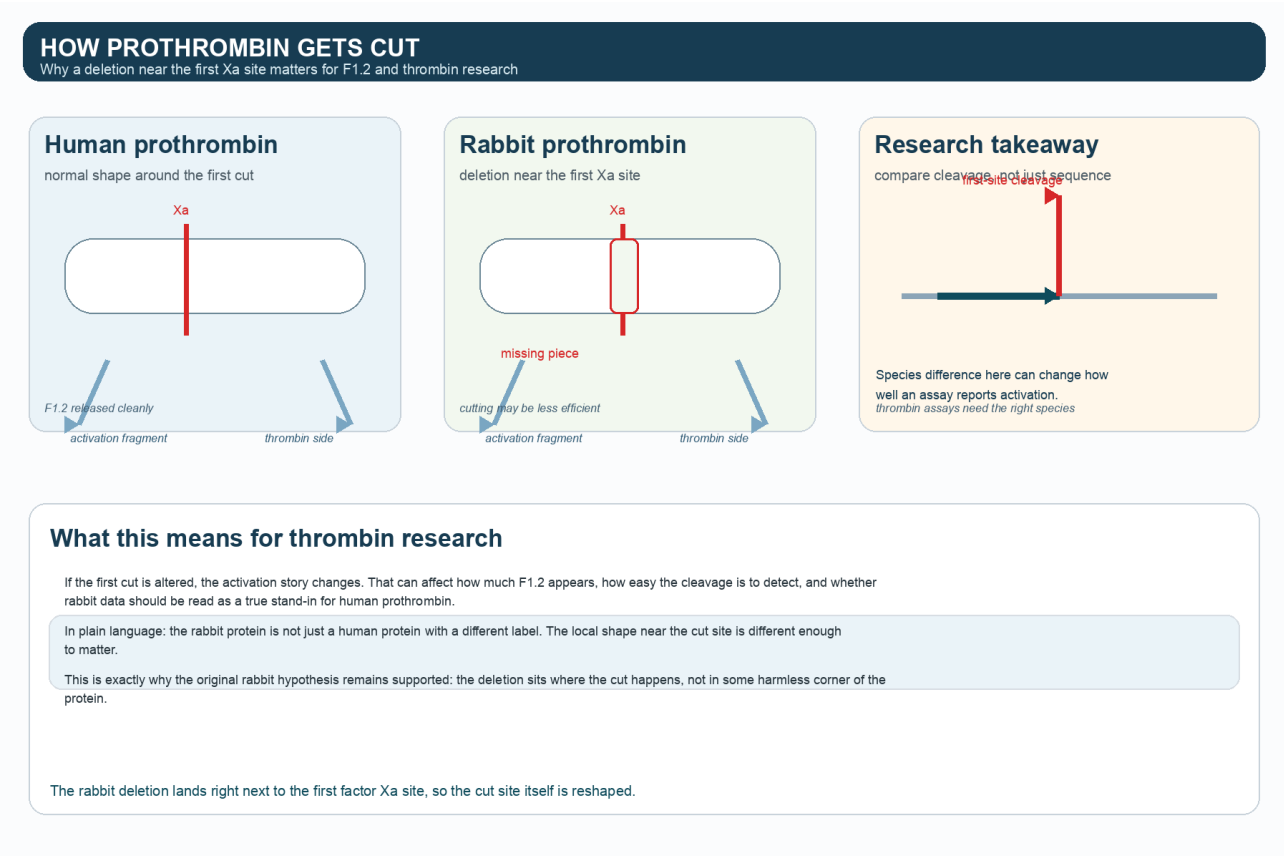
4. How to read the renderings

Think of each rendering as a different camera angle on the same story. You do not need to know chemistry to follow them. Start by looking for color and position, the way you would spot a red flag on a sports field or a missing brick in a wall.

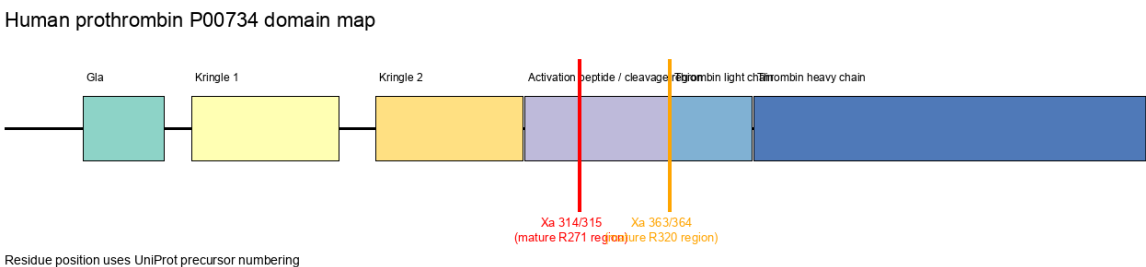
- Gray usually marks the human protein.
- Cyan usually marks the rabbit protein.
- Red marks the first Xa window, which is the most important region here.
- Orange marks the second Xa window, which helps us compare against a nearby control region.
- Magenta marks the deletion or gap region that makes rabbit different, like a missing stair in the middle of a staircase.

If the red region and the magenta region line up in a way that changes the shape of the protein, that supports the idea that rabbit prothrombin may not behave the same way as human prothrombin when the first cleavage happens. In other words, the key cutting site sits next to a shape change that could make the rabbit version harder to cut in the same way, and the 3D view makes that hard to miss.

5. Figure tour



Cleavage mechanism. This is the new magazine-style mechanism graphic. It shows how prothrombin gets cut, where F1.2 comes from, and why the rabbit deletion matters for thrombin research.



Domain map. This is the big-picture map. It shows the protein's parts and where the two cleavage sites live, like a blueprint for a machine.

Alignment window around the first factor Xa-sensitive site

	300										314										330																				
human	A	V	E	E	E	T	G	D	G	L	D	E	D	S	D	R	A	I	E	G	R	T	A	T	S	E	Y	Q	T	F	F	N	P	R	T	F	G	S	G	E	A
rabbit	A	I	E	E	E	-	-	-	-	-	V	L	G	L	E	E	A	I	E	G	R	T	T	E	Q	E	F	Q	T	F	F	N	Q	E	T	F	G	T	G	E	A

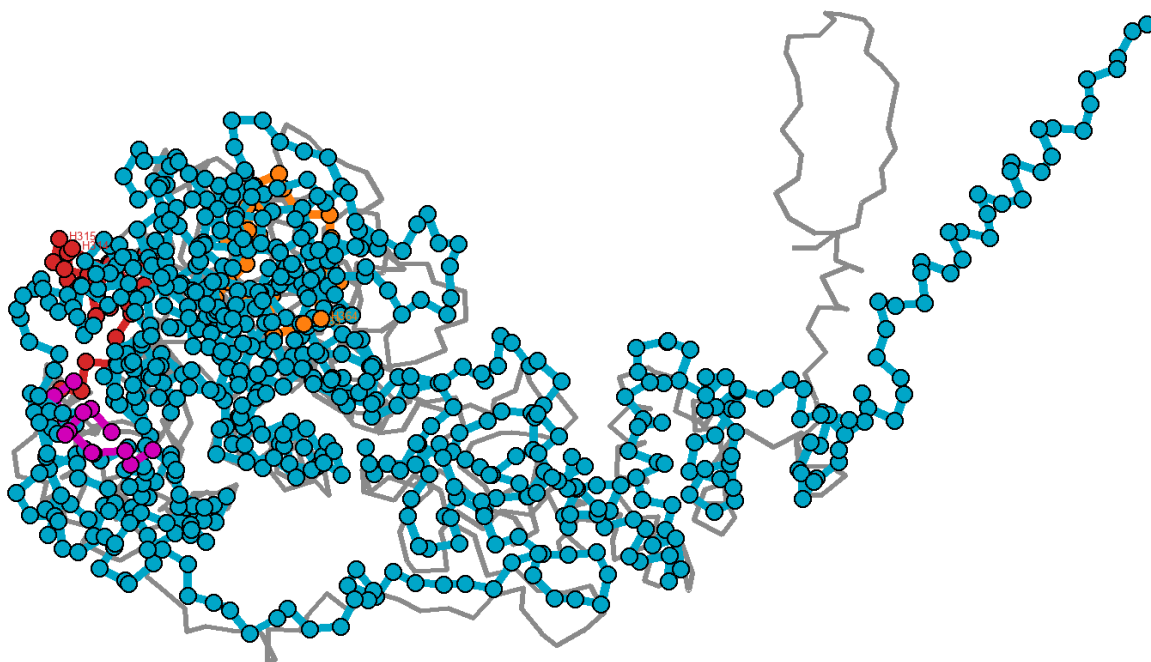
Magenta marks rabbit gaps/deletions; red marks human precursor Arg314 first Xa site.

Sequence alignment. This is the side-by-side sequence comparison around the first Xa site. The rabbit gap appears in magenta, like a missing word in a sentence that makes the whole line harder to read.

Whole prothrombin CA-trace overlay

Human and rabbit CA traces are projected after using the available aligned rabbit model.

- human CA trace
- rabbit CA trace
- first Xa window
- second Xa window
- deletion/gap region



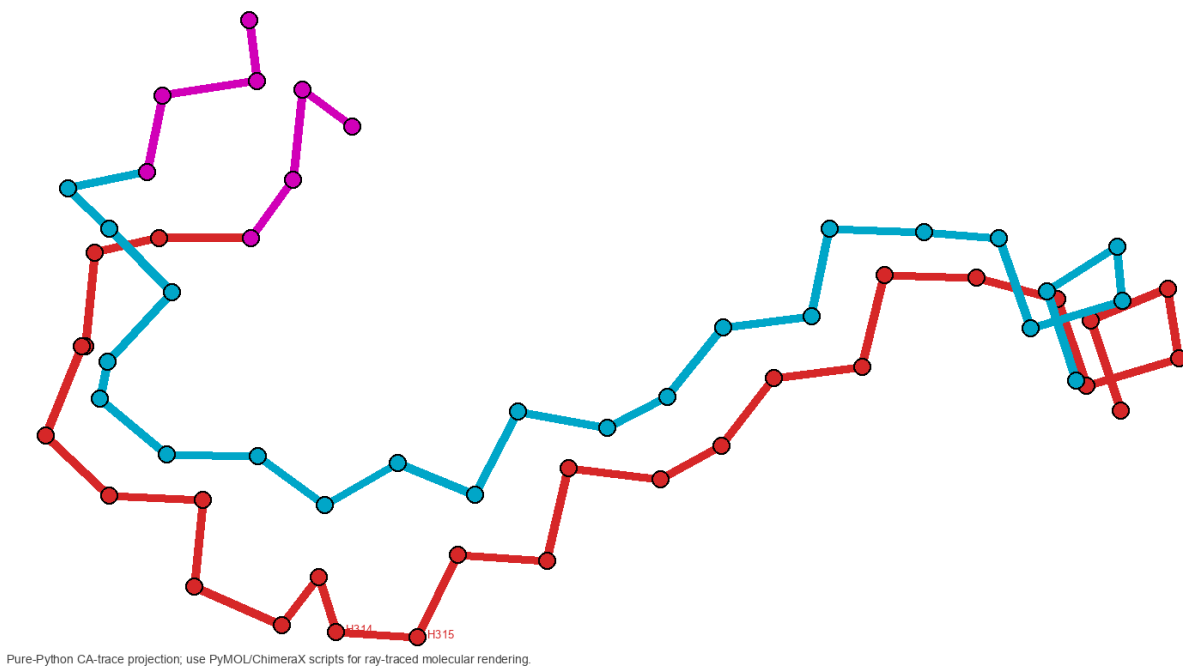
Pure-Python CA-trace projection; use PyMOU/ChimeraX scripts for ray-traced molecular rendering

Whole overlay. This 3D-style panel shows the full protein layout, with human and rabbit laid over each other like two transparencies, except one has a cut-out section near the important site.

First Xa cleavage-site closeup: human precursor 300-330

Human and rabbit CA traces are projected after using the available aligned rabbit model.

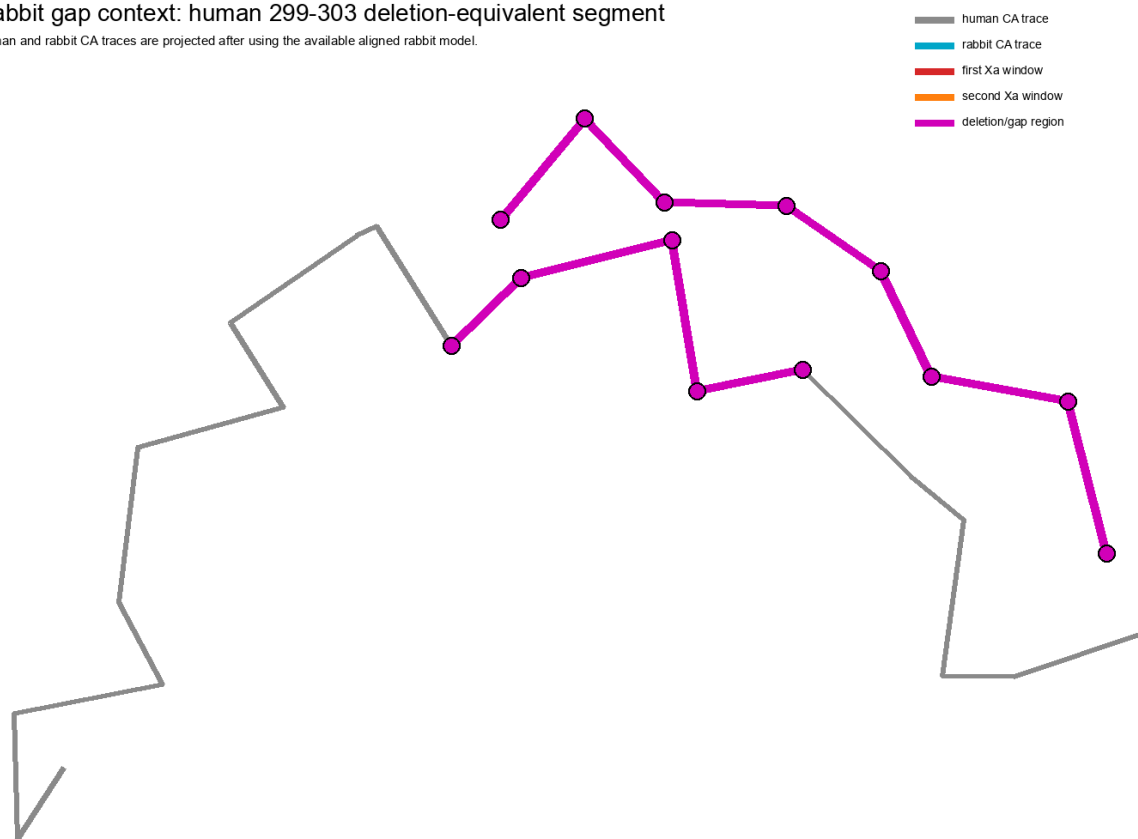
— human CA trace
— rabbit CA trace
— first Xa window
— second Xa window
— deletion/gap region



First Xa closeup. This zooms in on the key region near the first cleavage site, where the rabbit shape has a noticeable break in the pattern, like a zipper with one tooth missing.

Rabbit gap context: human 299-303 deletion-equivalent segment

Human and rabbit CA traces are projected after using the available aligned rabbit model.

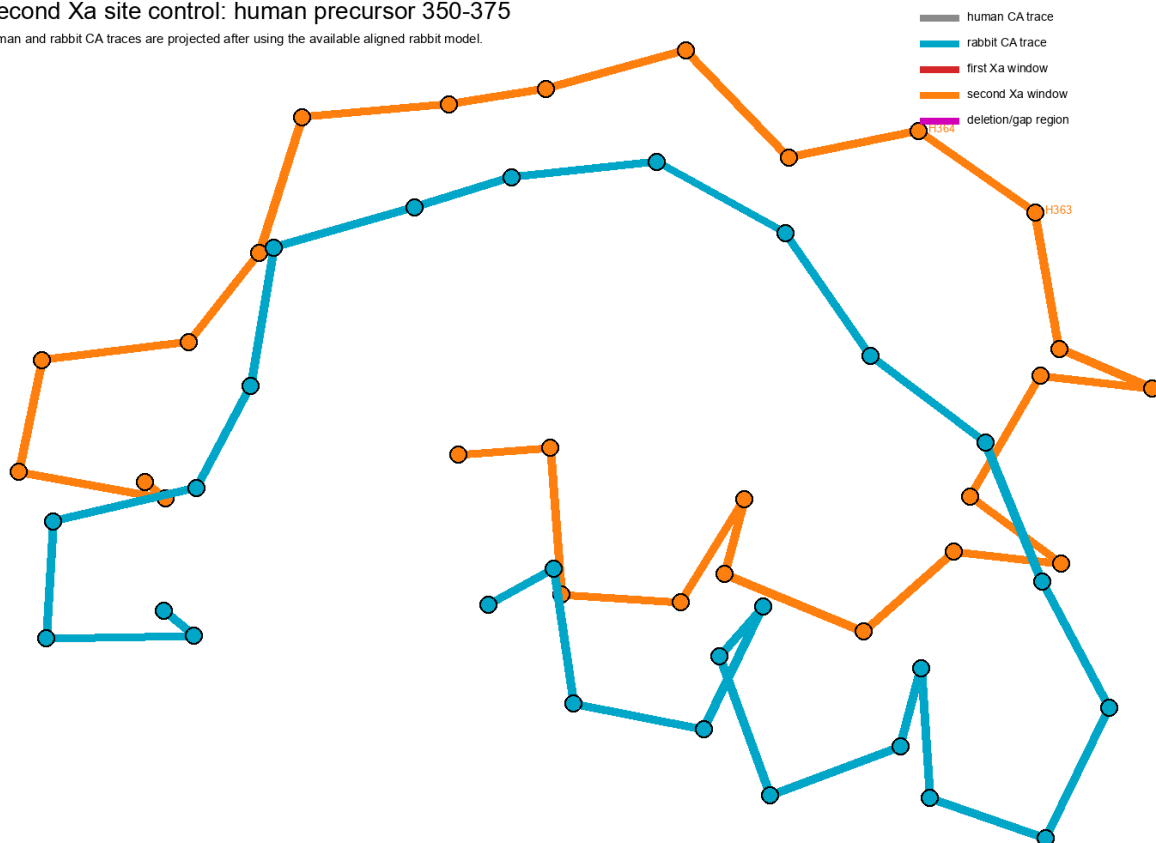


Pure-Python CA-trace projection; use PyMOU/ChimeraX scripts for ray-traced molecular rendering.

Deletion context. This is the most important picture for the hypothesis. It shows the rabbit deletion region in 3D context, like a notch cut out of one side of a keyhole so the key no longer fits the same way.

Second Xa site control: human precursor 350-375

Human and rabbit CA traces are projected after using the available aligned rabbit model.

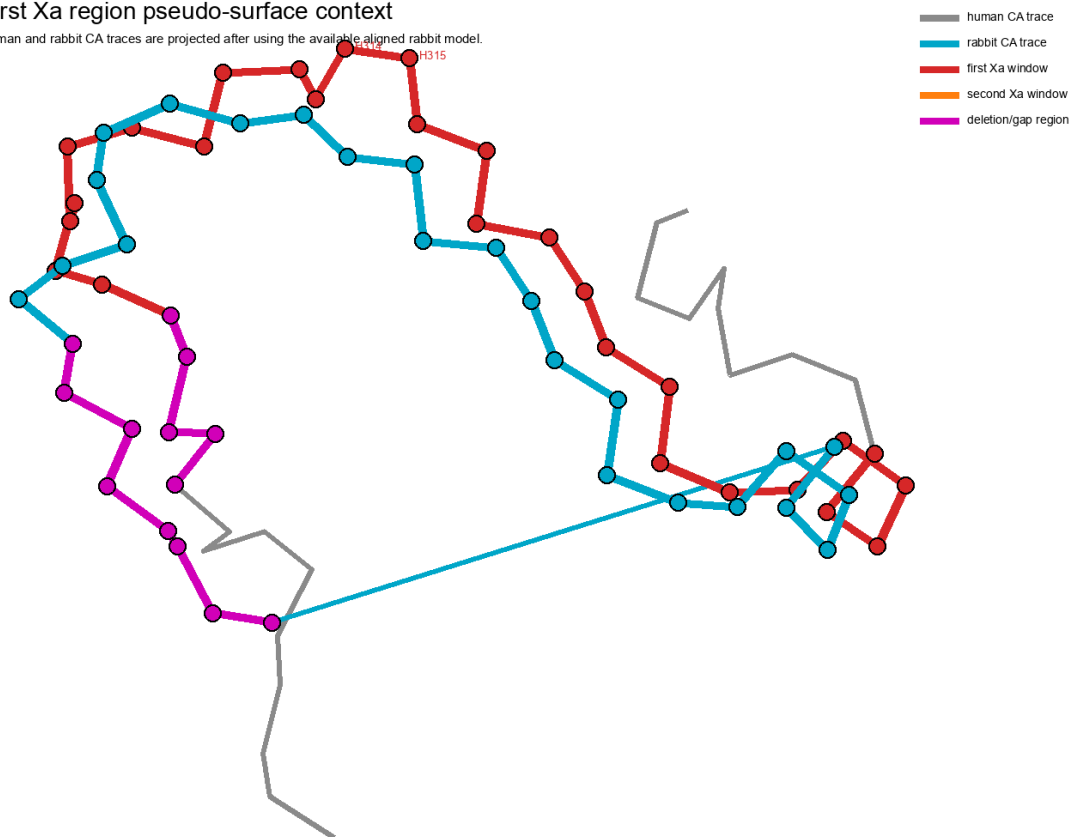


Pure-Python CA-trace projection; use PyMOU/ChimeraX scripts for ray-traced molecular rendering.

Second Xa control. This zooms in on the second cleavage site so you can compare it with the first and see that not every region is equally different.

First Xa region pseudo-surface context

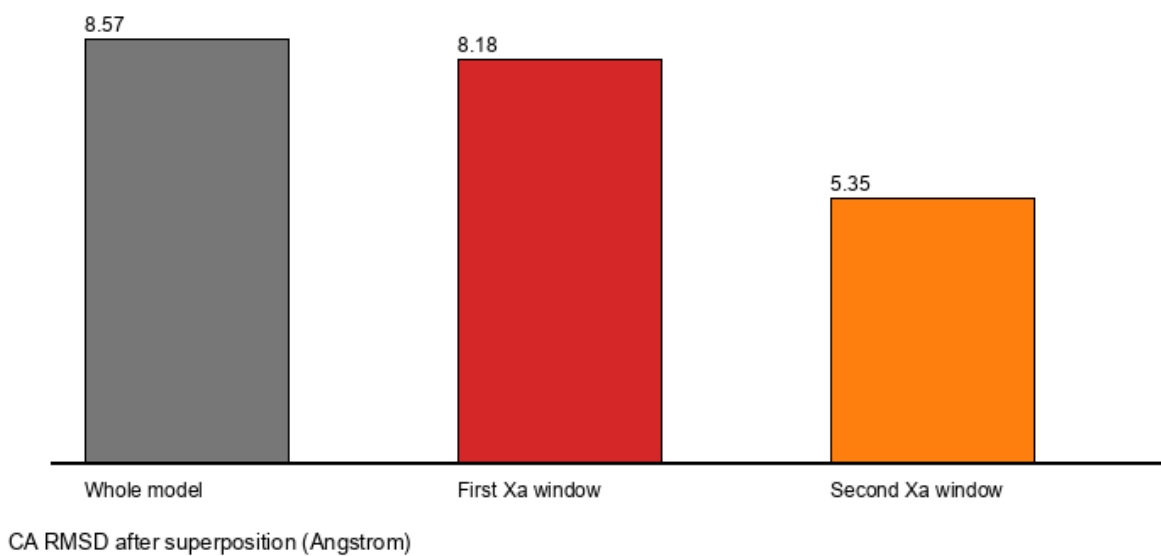
Human and rabbit CA traces are projected after using the available aligned rabbit model.



Pure-Python CA-trace projection; use PyMOU/ChimeraX scripts for ray-traced molecular rendering.

First Xa surface view. This view helps you see the first site on the surface of the protein, like a landing pad with a broken edge right where the plane needs to touch down.

Human vs rabbit prothrombin structural comparison



Metrics bar plot. This simple chart summarizes the RMSD metrics generated by the workflow.

6. Conclusion and the original hypothesis

The origin hypothesis was that rabbit prothrombin is not a clean model for the human F1.2 assay because the first cleavage region is different. The analysis strongly supports that idea. The rabbit sequence has a deletion near the human first Xa site, and the renderings show that difference in a way that is easy to see, like noticing that one shoe has a chunk missing exactly where the lace should hold tight.

That said, the project also keeps the scientific limit in view: a visual model cannot prove reaction speed. To know exactly how fast factor Xa cuts rabbit prothrombin, you would still need direct biochemical experiments with the prothrombinase complex.

Closing the loop: the sequence difference, the 3D mapping, and the old rabbit experiment all point in the same direction. The pictures do not just hint at the original hypothesis; they make it look very likely. The final answer about kinetics still belongs to the lab bench, but the structural story is already clear.

Current rendering status: complete_human_rabbit_overlay. Structure metric status: complete.

Bottom line: rabbit prothrombin is useful as a biological clue, but not as a direct copy of human prothrombin for first-site F1.2 assays. The 3D difference at the cleavage site makes that mismatch easy to picture and gives strong support to the original idea.

Appendix: files and reproducibility

Everything in this PDF comes from files that are already in the project. The renderings were generated with the local scripts, and the report text was derived from the same outputs.

Generated renderings included in this PDF

- outputs/renderings/01_whole_overlay.png
- outputs/renderings/02_first_xa_closeup.png
- outputs/renderings/03_deletion_flanks.png
- outputs/renderings/04_second_xa_control.png
- outputs/renderings/05_first_xa_surface.png

Short report source

```
# Human vs Rabbit Prothrombin Comparison
```

```
## 1. Question
```

```
Can rabbit prothrombin be used to model human prothrombin activation / F1.2 generation?
```

```
## 2. Background
```

```
PubMed PMID 10030826, "Prothrombin Activation in Rabbits," tested whether rabbit prothrombin activation fragment F1.
```

```
F1.2 assays depend on prothrombin activation chemistry: factor Xa cleavage of prothrombin, in the prothrombinase complex.
```

```
## 3. Sequence Comparison
```

- Human sequence: UniProt `P00734`.
- Rabbit sequence accession used: `XP\_002709128.3`.
- Human UniProt precursor numbering is used as the source of truth.
- First Xa site label: precursor `314/315`, equivalent to the mature/literature `R271` region.
- Second Xa site label: precursor `363/364`, equivalent to the mature/literature `R320` region.
- First-site-proximal deletion detected: `True`.

```
Alignment excerpt around the first Xa-sensitive region:
```

```
```text
human_pos: 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320
human: A V E E E T G D G L D E D S D R A I E G R T A T S E Y
rabbit: A I E E E - - - - - V L G L E E A I E G R T T E Q E F
```

...

See `data/alignments/human\_rabbit\_global\_alignment.csv` and `outputs/deletion\_summary.csv` for exact mapped position

#### ## 4. Structure Comparison

- Human structure source: `AlphaFold DB` [https://alphafold.ebi.ac.uk/files/AF-P00734-F1-model\\_v6.pdb](https://alphafold.ebi.ac.uk/files/AF-P00734-F1-model_v6.pdb).
- Rabbit model source: `AlphaFold website prediction imported locally`.
- Rabbit model path/status: `data/structures/rabbit\_AF\_model.pdb`.

The rabbit model was imported from the AlphaFold website output and saved as `data/structures/rabbit\_AF\_model.pdb`.

Structure metrics:

```
```json
{
  "status": "complete",
  "rabbit_model_type": "alphafold_website_prediction",
  "rabbit_model_source": "AlphaFold website prediction imported locally",
  "superposition_ca_pairs": 616,
  "whole_model_ca_rmsd_angstrom": 8.571601429967375,
  "whole_model_ca_pairs": 616,
  "first_xa_window_300_330_ca_rmsd_angstrom": 8.177920729083395,
  "first_xa_window_ca_pairs": 27,
  "second_xa_window_350_375_ca_rmsd_angstrom": 5.348764487236,
  "second_xa_window_ca_pairs": 26,
  "first_site_gap_human_positions": [
    299,
    300,
    301,
    302,
    303
  ],
  "note": "AlphaFold-derived structures are hypotheses and do not establish cleavage kinetics."
}
```
```

The structural analysis superimposes rabbit onto human using aligned CA atoms when both models are available. Local

#### ## 5. 3D Rendering Plan and Outputs

- Rabbit model status: `aligned`.
- Rabbit model source type: `alphafold\_website\_prediction`.
- Pure-Python rendering status: `complete\_human\_rabbit\_overlay`.
- Auto-render status: `skipped\_no\_renderer`.
- Human deletion/gap positions highlighted: `[299, 300, 301, 302, 303]`.
- Human deletion/gap sequence highlighted: `TGDGL`.
- Render scripts: `outputs/render\_human\_rabbit\_prothrombin.pml` and `outputs/render\_human\_rabbit\_prothrombin.cxc`.
- Generated renderings:
  - `outputs/renderings/01\_whole\_overlay.png`
  - `outputs/renderings/02\_first\_xa\_closeup.png`
  - `outputs/renderings/03\_deletion\_flanks.png`
  - `outputs/renderings/04\_second\_xa\_control.png`
  - `outputs/renderings/05\_first\_xa\_surface.png`

These renderings are intended to make the first-site-proximal alignment gap visible in structural context. Pure-Pyth

#### ## 6. Biological Interpretation

Human prothrombin efficiently generates F1.2 after activation, whereas PMID 10030826 reports that rabbit plasma does

#### ## 7. Limitations

- AlphaFold and ColabFold models are structural hypotheses, not experimental proof.
- Prothrombinase biology includes factor Xa, factor Va, phospholipid membrane, and Ca<sup>2+</sup>.
- Gla-domain membrane binding and activation-complex dynamics are difficult to infer from static monomer structures.
- Kinetics and prothrombinase complex biology require experimental validation, such as biochemical cleavage assays a

#### ## 8. Files Generated

- `data/alignments/human\_rabbit\_global\_alignment.csv`
- `data/alignments/human\_rabbit\_global\_alignment.txt`
- `data/raw/ncbi\_search\_rabbit.json`
- `data/raw/uniprot\_search\_rabbit.json`

- `data/sequences/human\_P00734.fasta`
- `data/sequences/rabbit\_F2.fasta`
- `data/structures/RUN\_COLABFOLD\_INSTRUCTIONS.md`
- `data/structures/human\_AF\_P00734.pdb`
- `data/structures/rabbit\_AF\_model.pdb`
- `data/structures/rabbit\_aligned\_to\_human.pdb`
- `data/structures/rabbit\_colabfold\_input.fasta`
- `data/structures/rabbit\_threaded\_from\_human\_AF\_P00734.pdb`
- `outputs/deletion\_summary.csv`
- `outputs/deletion\_summary.txt`
- `outputs/figures/cleavage\_alignment.png`
- `outputs/figures/domain\_map.png`
- `outputs/figures/metrics\_barplot.png`
- `outputs/pnas\_style\_report.md`
- `outputs/render\_human\_rabbit\_prothrombin.cxc`
- `outputs/render\_human\_rabbit\_prothrombin.pml`
- `outputs/rendering\_manifest.json`
- `outputs/renderings/01\_whole\_overlay.png`
- `outputs/renderings/02\_first\_xa\_closeup.png`
- `outputs/renderings/03\_deletion\_flanks.png`
- `outputs/renderings/04\_second\_xa\_control.png`
- `outputs/renderings/05\_first\_xa\_surface.png`
- `outputs/report.md`
- `outputs/report.pdf`
- `outputs/scientific\_american\_style\_explainer.pdf`
- `outputs/structure\_metrics.csv`
- `outputs/structure\_metrics.json`
- `outputs/view\_human\_rabbit\_prothrombin.pml`